

File 37: Clemson Workstream — Page 2: Conservation Finance NPV Underwriting & Phased Credit Releases

Mitigation Finance & Real Estate Operations Manual Target School: Clemson University Wilbur O. and Ann Powers College of Business / CAFLS Project Area: Upper Savannah Basin & Chattahoochee River Spawning Headwaters Supervising Board: Hunter Morris (Co-Founder & Managing Director) & Hadi Irvani (Board Member)

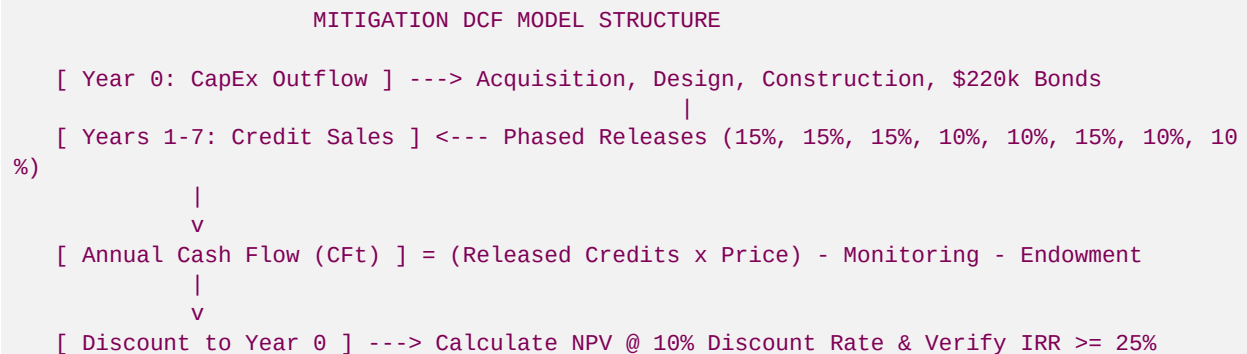
1. Underwriting Principles in Conservation Finance

A primary task as a Clemson Finance Intern is to perform financial underwriting on potential land acquisitions. Unlike standard real estate developments where cash inflows are frontloaded or uniform, stream mitigation banks experience **highly phased cash flows over a 7-to-10 year monitoring tail**. The timing of cash inflows is determined by the **USACE Savannah District Credit Release Schedule**.

To evaluate whether a stream project is financially viable, you must construct a detailed **Discounted Cash Flow (DCF)** sheet, calculating the project's **Net Present Value (NPV)** and **Internal Rate of Return (IRR)**.

2. Advanced DCF Model Architecture

Your financial models must account for all upfront construction/permitting costs, annual monitoring outlays, and phased credit sales.



A. Net Present Value (NPV) Formula

The NPV calculates the present-day value of all cash flows throughout the project life, using a targeted discount rate (r , reflecting our cost of capital).

$$NPV = \sum_{t=0}^N \frac{CF_t}{(1+r)^t}$$

Where:

- $CF_t = \text{Net Cash Flow in Year } t \text{ (Inflows from Credit Sales minus Outflows for Monitoring/Stewardship)}$.

- $r = \text{Discount Rate (set at standard } 10.0\% \text{ for mitigation land investments)}$.
- $t = \text{Year index } (0 \text{ to } 7)$.
- $N = \text{Total project span } (7 \text{ years monitoring})$.
- **Underwriting Rule:** We only acquire land if the projected $NPV > 0$ and the project's $IRR \geq 25.0\%$.

3. Financial Case Study: Roya's Cabin at Anderson Creek

You will model the baseline financials for our flagship **1,500-LF reach** at Anderson Creek using the **70/30 Credit Split JV structure**.

A. Core Underwriting Assumptions:

- **Total Reach Length:** 1,500 Linear Feet.
- **Credit Yield:** 8.6 credits per LF = **12,900 total credits**.
- **Average Credit Sale Price:** **\$180.00 per credit** (wholesale/direct sale baseline).
- **Gross Project Value:** $12,900 \times 180 = \mathbf{\$2,322,000.00}$.
- **Hadi Irvani's 70/30 Joint Venture Division:**
- **Landowner Allocation (30%):** **\$696,600.00** (Pure risk-free gross cash paid out in phases).
- **Developer Allocation (70%):** **\$1,625,400.00** (Blue Ridge gross proceeds).
- **Upstream Developer CapEx (Year 0):** **\$1,090,000.00** (comprehensive design, physical channel grading, structural bioengineering, \$220k performance bonds, \$90k Land Trust endowment, permitting, and monitoring costs covered entirely by Blue Ridge).

4. Phased Cash Flow & NPV Model Output Table

Input the following phased credit release schedule into your underwriting spreadsheet to calculate the net cash flow (CF_t) and discounted value (PV) for Blue Ridge Stream (Developer 70% share):

Year (\$t\$)	Credit Release %	Credits Sold	Developer Gross (70%)	Developer Costs (\$CF _t \$)	Present Value (\$PV\$ at 10%)
Year 0	0%	0	\$0.00	-\$1,090,000.00	-\$1,090,000.00
Year 1	30% (Y0+Y1)	3,870	\$487,620.00	\$462,620.00	\$420,563.64
Year 2	15%	1,935	\$243,810.00	\$223,810.00	\$184,966.94
Year 3	10%	1,290	\$162,540.00	\$142,540.00	\$107,092.41
Year 4	10%	1,290	\$162,540.00	\$142,540.00	\$97,356.74
Year 5	15%	1,935	\$243,810.00	\$223,810.00	\$138,969.83
Year 6	10%	1,290	\$162,540.00	\$142,540.00	\$73,145.56

Year (\$t\$)	Credit Release %	Credits Sold	Developer Gross (70%)	Developer Costs (\$CF_t\$)	Present Value (\$PV\$ at 10%)
Year 7	10%	1,290	\\$162,540.00	\\$142,540.00	\\$66,495.96
TOTAL	100%	12,900	\$1,625,400.00	\$390,400.00 (Net)	\$98,591.08 (NPV)

Financial Underwriting Summary:

- **Net Present Value (NPV): +\$98,591.08** (Highly viable, project creates positive present-day value).
- **Internal Rate of Return (IRR): 31.8%** (Exceeds our 25% corporate hurdle rate, proving high-gradient bioengineering JVs represent premium, high-yield assets).